VERDANTTECH
SOLUTIONS
CEA OPERATIONS CONSULTING

CEA Operations and Economics Review

INDEPENDENT OPERATIONAL DUE DILIGENCE · PRE-INVESTMENT THROUGH ACTIVE ASSET MANAGEMENT

CEA operations are complex enough that standard due diligence doesn't always surface the right questions. The numbers in the model may be honest — and still not reflect what the facility will actually do at production scale.

CEA pro formas follow predictable patterns: yield assumptions that don't account for ramp period, energy models that underestimate consumption at production density, labor cost structures built on efficiency curves that take years to develop. These aren't unusual errors — they're structural features of how facilities get modeled before they operate. Capital gets committed against projections built on assumed inputs, and the gap between projection and performance compounds silently until it becomes a capital problem.

The question for an investor or lender isn't whether the technology works or whether the market exists. It's whether the operational assumptions in a specific plan will hold — and that requires someone who has read those assumptions against actual operating performance.

My job is to tell you what the model is actually assuming — and whether those assumptions are defensible.

WHERE CEA MODELS TYPICALLY BREAK DOWN

YIELD & OUTPUT

- Capacity modeling
- Crop yield assumptions
- Throughput validation

COST STRUCTURE

- Energy use
- Labor efficiency curves
- Supply chain unit costs

TECHNOLOGY & INTEGRATION

- System selection assumptions
- Production/financial sensitivity
- Operational integration plans

WHERE I CAN HELP

PRE-INVESTMENT ANALYSIS · FOR INVESTORS AND LENDERS EVALUATING A CEA OPPORTUNITY BEFORE CAPITAL IS COMMITTED

PRO FORMA STRESS-TESTING

I identify which assumptions are defensible against actual operating experience — and which aren't.

- Yield, throughput, and ramp period modeling
- Energy sensitivity and labor cost structure
- Assumptions the facility will struggle to meet

FEASIBILITY & PRE-DEVELOPMENT REVIEW

For facilities still in design: does the proposed system actually produce the economics the model projects?

- Crop, scale, and technology stack fit
- Energy load modeling and throughput analysis
- Pre-commitment gap identification

POST-INVESTMENT / ACTIVE ASSET MANAGEMENT · FOR INVESTORS AND LENDERS MANAGING A FACILITY ALREADY IN OPERATION

OPERATIONS ASSESSMENT

A structured review of whether the farm is performing against its model — built around your interests.

- Yield per square foot vs. projection
- Actual vs. projected labor cost and energy consumption
- Food safety compliance status and unit economics

UNIT ECONOMICS REVIEW

Where are the gaps between projected and actual performance coming from — and which are fixable?

- Structural vs. solvable problem identification
- Realistic improvement range for this specific facility
- Direct path to operational improvement engagement if needed

BOTH PHASES

TECHNOLOGY & SYSTEMS REVIEW

Independent evaluation of the facility's technology stack — relevant whether the facility is still in design or already operating.

- Growing systems, climate & irrigation, harvest and packaging technology
- Food safety management, data collection, and general automation
- Is the system designed to run at projected throughput? Are there known integration or reliability issues in the chosen configuration?

BACKGROUND & EXPERIENCE

I've spent over a decade in the CEA industry at Vertical Harvest. I started this part of my career inside their small-scale multi-crop facility in Jackson, Wyoming — first as a founding board member and general business consultant, then joining the company full-time in 2016 as a key executive alongside the co-founders.

Like leaders in most startups, I held many roles over time. As CFO I provided financial leadership and gained a deep understanding of the critical cost drivers and unit economics of the business. As Director of Farm Systems I served as technical systems lead, where I led the re-think, re-design, and upgrade of existing operational and growing systems.

In my final role as Chief of Farm Systems, I served as the operator-side systems lead for the design and build-out of their large-scale automated facility in Maine — one of the more highly automated CEA facilities built in North America, designed for space and labor efficiency. I coordinated expert vendor and internal teams across every major discipline. I was responsible for all final equipment and software purchase selections, and had design oversight for how all the pieces fit together operationally. Throughout, I built and applied production economics models for the live operation to understand and project unit costs, yield prediction, process flow analysis, and projection-to-actual tracking.

VerdantTech Solutions is the independent consulting practice built on that background.

SCOPE AND ENGAGEMENT

My engagement scope is the operational and unit economics picture — sales strategy, pricing, and market development are outside it. Operational improvement work is a separate engagement, but one I can take on directly. For investors who want operational support built in as a condition of the investment, that structure is available.

I will disclose all existing vendor and operator relationships to make sure any conflicts of interest are clear before any engagement begins.

If you're evaluating a CEA opportunity and want an operator-side read on the assumptions — the right next step is a conversation.

LET'S TALK

EMAIL
arik@verdant.biz

PHONE
(307) 413-4104

WEB
verdant.biz